EXPERIMENT 2: UPDATE DEFENDER DEFINITIONS AND RUN A QUICK SCAN

Task: T5

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1. Aim

To develop and execute a Bash shell script using ClamAV to update virus definitions, recursively scan a specified directory, display the scan date, infected file count and scan duration, and append the scan summary to a log file. Windows Defender does not exist on Kali Linux, Ubuntu or macOS, so this ClamAV script is the substitute for the Defender update-and-quick-scan task on those systems.

2. Course outcome and module

Course Outcome: CO1 – Configure and manage system protection tools and evaluate their ability to defend against threats.

Module: System Protection and Threat Prevention.

3. Tools and versions

Tool	Exact version run	Operating system
ClamAV (`freshclam`, `clamscan`)	ClamAV (`clamscan --version`)	Kali Linux (version used)
Bash	GNU Bash (`bash --version`)	Kali Linux (version used)
GNU nano	GNU nano (`nano --version`)	Kali Linux (version used)

4. Program or commands

```
#!/bin/bash

FOLDER="$1"
LOG_FILE="$HOME/clamav_exp/av_scan.log"

if [ -z "$FOLDER" ]; then
    echo "Usage: bash av_scan.sh <folder>"
    exit 1
fi

if [ ! -d "$FOLDER" ]; then
    echo "Error: Folder does not exist!"
    exit 1
fi

SCAN_DATE=$(date)
START_TIME=$(date +%s)

echo "Starting ClamAV scan..."

SCAN_OUTPUT=$(clamscan -r -i "$FOLDER" 2>&1)
SCAN_STATUS=$?

END_TIME=$(date +%s)
SCAN_TIME=$((END_TIME - START_TIME))

INFECTED=$(echo "$SCAN_OUTPUT" |
    awk '/^Infected files:/ {print $3}')

echo
echo "------ SCAN SUMMARY ------"
echo "Scan Date: $SCAN_DATE"
echo "Scanned Folder: $FOLDER"
echo "Infected Files: $INFECTED"
echo "Scan Time: $SCAN_TIME seconds"
echo "--------------------------"

{
    echo "------ SCAN SUMMARY ------"
```

```
    echo "Scan Date: $SCAN_DATE"
    echo "Scanned Folder: $FOLDER"
    echo "Infected Files: $INFECTED"
    echo "Scan Time: $SCAN_TIME seconds"
    echo "--------------------------"
} >> "$LOG_FILE"

echo "Results appended to $LOG_FILE"
```

5. How to run

1. Update the virus definitions

```
sudo freshclam
```

2. Create the script

```
nano av_scan.sh
```

3. Run the script on a folder

```
bash av_scan.sh "$HOME/clamav_exp/test_folder"
```

4. View the log

```
cat ~/clamav_exp/av_scan.log
```

The same four commands are used on Ubuntu, Kali Linux and macOS.

6. Sample output with screenshots

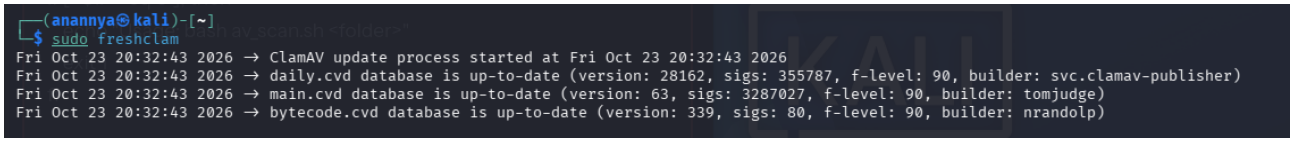


Figure T05-1: ClamAV Definition Update

The freshclam command updates the ClamAV virus definitions to ensure the scanner uses the latest malware signatures.

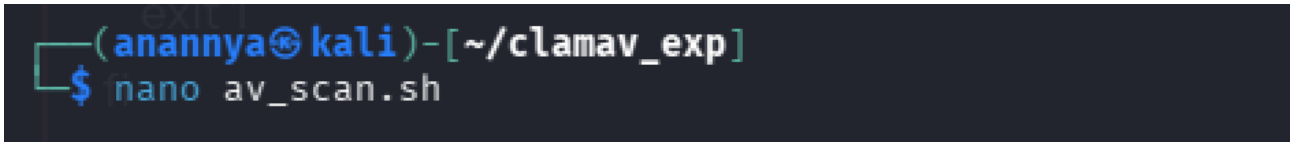


Figure T05-2: Script Creation

The av_scan.sh script is created and edited using the GNU nano text editor.

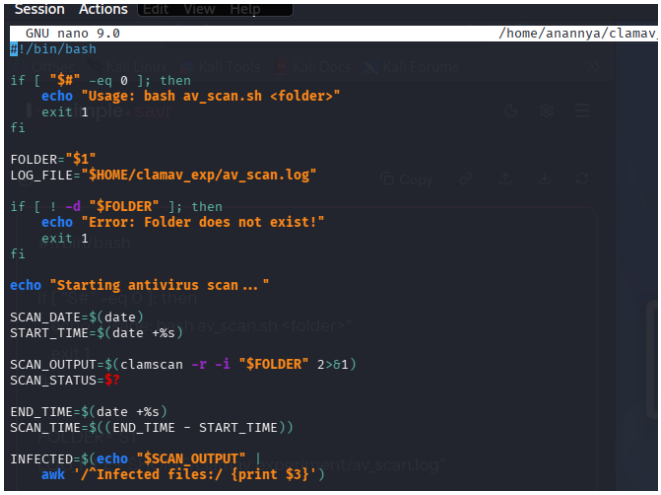


Figure T05-3: Script Validation and Scanning

The script validates the folder argument, records the scan start time, and performs a recursive ClamAV scan.

```
if [ -z "$INFECTED" ]; then
    echo "$SCAN_OUTPUT"
    echo "Unable to determine infection count."
    exit 1
fi

echo
echo "──── SCAN SUMMARY ────"
echo "Scan Date: $SCAN_DATE"
echo "Scanned Folder: $FOLDER"
echo "Infected Files: $INFECTED"
echo "Scan Time: $SCAN_TIME seconds"
echo "──────────────────────"

{
    echo "──── SCAN SUMMARY ────"
    echo "Scan Date: $SCAN_DATE"
    echo "Scanned Folder: $FOLDER"
    echo "Infected Files: $INFECTED"
    echo "Scan Time: $SCAN_TIME seconds"
    echo "──────────────────────"
} >> "$LOG_FILE"

echo "Results appended to $LOG_FILE"
```

Figure T05-4: Scan Summary and Logging

The script calculates the scan duration, extracts the infected file count, displays the summary, and appends it to the log file.

```
┌──(anannya㉿kali)-[~/clamav_exp]
└─$ bash ~/clamav_exp/av_scan.sh "$HOME/clamav_exp/test_folder"
Starting antivirus scan...

──── SCAN SUMMARY ────
Scan Date: Friday 23 October 2026 09:01:31 PM IST
Scanned Folder: /home/anannya/clamav_exp/test_folder
Infected Files: 0
Scan Time: 5 seconds
──────────────────────
Results appended to /home/anannya/clamav_exp/av_scan.log
```

Figure T05-5: ClamAV Scan Output

The terminal displays the scan date, scanned folder, infected file count, and scan duration.

```
┌──(anannya㉿kali)-[~/clamav_exp]
└─$ cat ~/clamav_exp/av_scan.log
──── SCAN SUMMARY ────
Scan Date: Friday 23 October 2026 09:01:31 PM IST
Scanned Folder: /home/anannya/clamav_exp/test_folder
Infected Files: 0
Scan Time: 5 seconds
──────────────────────
```

Figure T05-6: Scan Log

The log file displays the scan summary saved by the script.

7. Result

The ClamAV script was successfully executed on Kali Linux. It scanned the specified folder recursively, displayed the scan date, infected file count and scan duration, and appended the scan summary to the log file.

8. How this differs from others

Aspect	This version (T5)	Windows Defender version
Platform	Kali Linux, Ubuntu or macOS	Windows
Scanner	ClamAV (`clamscan`)	Windows Defender
Definitions update	`freshclam`	Defender definition update
Scan	Recursive scan of a specified folder	Quick scan
Result	Date, infected count and scan duration recorded in a log	Last scan date and result summary

9. Viva questions and answers

1. What is ClamAV?

ClamAV is an open-source antivirus engine used to detect malware and viruses.

2. What do the `-r` and `-i` options of `clamscan` do?

The `-r` option scans directories and their subdirectories recursively, while `-i` displays only infected files.

3. Why is the folder path passed as an argument?

It allows the user to specify the directory to be scanned when executing the script.

4. What is the purpose of `awk` in the program?

It extracts the infected file count from the ClamAV scan output.

5. Why is the log file opened using »?

The » operator appends new scan results without overwriting the existing log file.

10. Common Errors: Message, Cause, and Fix

Message	Cause	Fix
`Usage: bash av_scan.sh <folder>`	No folder argument provided	Pass the target folder while executing the script.
`Error: Folder does not exist!`	Invalid directory path	Check the directory name and path.
Permission denied	Insufficient access permission	Check the permissions of the target directory.
Database outdated	ClamAV signatures are outdated	Update the database using `freshclam`.

Declaration: AI tools were utilized for grammatical review, comparison with alternative versions, and assistance in preparing, reviewing, and verifying viva answers.